

If-Else Statements in JavaScript

What is an If-Else Statement?

An `if` statement is used to run a block of code only if a specified condition is true. You can use `else` or `else if` to run different blocks of code based on different conditions.

Basic Syntax

```
if (condition) {  
    // code to run if condition is true  
} else {  
    // code to run if condition is false  
}
```

Example

```
let age = 18;  
  
if (age >= 18) {  
    console.log("You are an adult.");  
} else {  
    console.log("You are a minor.");  
}
```

if-else-if Ladder

You can check multiple conditions using `else if`:

```
let score = 85;

if (score >= 90) {
  console.log("Grade: A");
} else if (score >= 80) {
  console.log("Grade: B");
} else if (score >= 70) {
  console.log("Grade: C");
} else {
  console.log("Grade: F");
}
```

Nested if Statements

You can place one `if` statement inside another:

```
let age = 25;
let hasID = true;

if (age >= 18) {
  if (hasID) {
    console.log("Access granted.");
  } else {
    console.log("ID required.");
  }
} else {
  console.log("Access denied. You must be at least 18.");
}
```

Using the Ternary Operator (Short Form)

The ternary operator is a shorter way to write simple `if-else` statements:

```
let isLoggedIn = true;

let message = isLoggedIn ? "Welcome back!" : "Please log in.";
console.log(message);
```

Summary

- Use `if` to test a condition.
- Use `else` for an alternative block if the condition is false.
- Use `else if` to test multiple conditions.
- The ternary operator is a compact way to write simple `if-else` logic.